INTERNATIONAL ORIENTEERING FEDERATION

Timing Manual

For World Orienteering Championships and Orienteering World Cup

Issued by IOF Office & IOF IT Commission

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1 Introduction to the Timing Manual

The TV-rights for World Orienteering Championships and Orienteering World Cup are nowadays sold to the National TV stations in several countries. Delivering a product to premiere customers comes with obligations to deliver consistent quality in every TV Broadcasted race. Therefore the IOF now introduces a Timing Manual, for Foot Orienteering, where the requirements connected to timekeeping are gathered.

This document describes the requirements, and ways to fulfil the requirements for timekeeping and integration with TV production, so that we can deliver a consistent quality of timekeeping and results in our TV broadcasts. The TV producer is entirely dependent on the output from the timekeeping team, and the National TV stations expects nothing but functioning timekeeping and results - always - with no exceptions.

In this Timing Manual the IOF present the requirements and possible technical solutions, so the risk of failure is minimised, and we can present our sport in a professional way, as expected by the National TV stations and viewers. The IOF also want to prevent discussions due to viewers getting the impression of wrong times due to how the pictures and result graphics are synchronised, or how people wear the transponder crossing the finish line.

The target group for the Timing Manual is professionals that will provide timekeeping and results services at World Orienteering Championships and Orienteering World Cup races, and organisers that need the requirements of timekeeping in order to set up their own organisation and get a tender for these services.

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1.0 Summary

To ensure consistent quality of televised World Orienteering Championships and World Cup events we (re)introduce requirements for higher accuracy in the timekeeping.

- Start gate at start, recording and transmitting the exact start time
- Photocells as a required system in interval start races, together with transponders for identification and backup timing
- Transponders and photo finish for first-to-finish-races
- Mandatory to display results from Sprint Relay and Knock-out Sprint during World Orienteering Championships and Orienteering World Cup 2023 with 1/10 precision.
- Organisers shall provide the primary transponder in all televised World Orienteering Championship and Orienteering World Cup races.

2 Categorisation of events

IOF Events are categorised into three levels. *Only* level 1 is in scope in this manual.

Events (examples)	Level	Precision in the result list
WOC and Orienteering World Cup with broad-cast TV	Level 1	1 sec precision in interval start races, operated with an accuracy of 1/10 sec. 1/10 sec precision in first-to-finish races.
All other high-level events, like WSOC, JWOC, ROC	Level 2	1 sec precision, electronic finish ("the athlete shall not need to stop at the fin-ish line, but run, cycle, ski across")
WRE, WMOC	Level 3	1 sec precision, finish punch allowed with any approved system solution

3 Requirements from the TV producer

This is a summary of the requirements from the tv producer:

1. Max latency for split-times and finish times: 500 milliseconds
2. Max latency for pre-warnings: 2 seconds
3. Use cabled connections, as these are more reliable and have lower latency.
    - If not possible, a dedicated radio link with a verified and consistent latency may be used.
4. Fully redundant backup in the finish (System A & System B) – and procedures for handling missing times
    - If a time is missing in system A, there must be a routine to get the time from system B within 30 seconds.
5. If the split times are not working, a procedure for backup, that is also connected to the TV graphics systems, must be in place.
6. Establish a routine for sending a photo finish still picture to the TV production quickly (within 30 seconds)
7. The infosystem for commentators must have access to all pre-warnings and live times, both for onsite and off-tube (remote) commentators

4 Timekeeping requirements

4.0 Determination of times

All races require correct times, but races that are produced for broadcast TV needs to show the correct time immediately, even when there are technical problems.

4.0.0 Individual finals

For interval-start races, we want the accuracy to be an order of magnitude better than the precision, i.e 1/10 second accuracy for times which are truncated to one second precision.

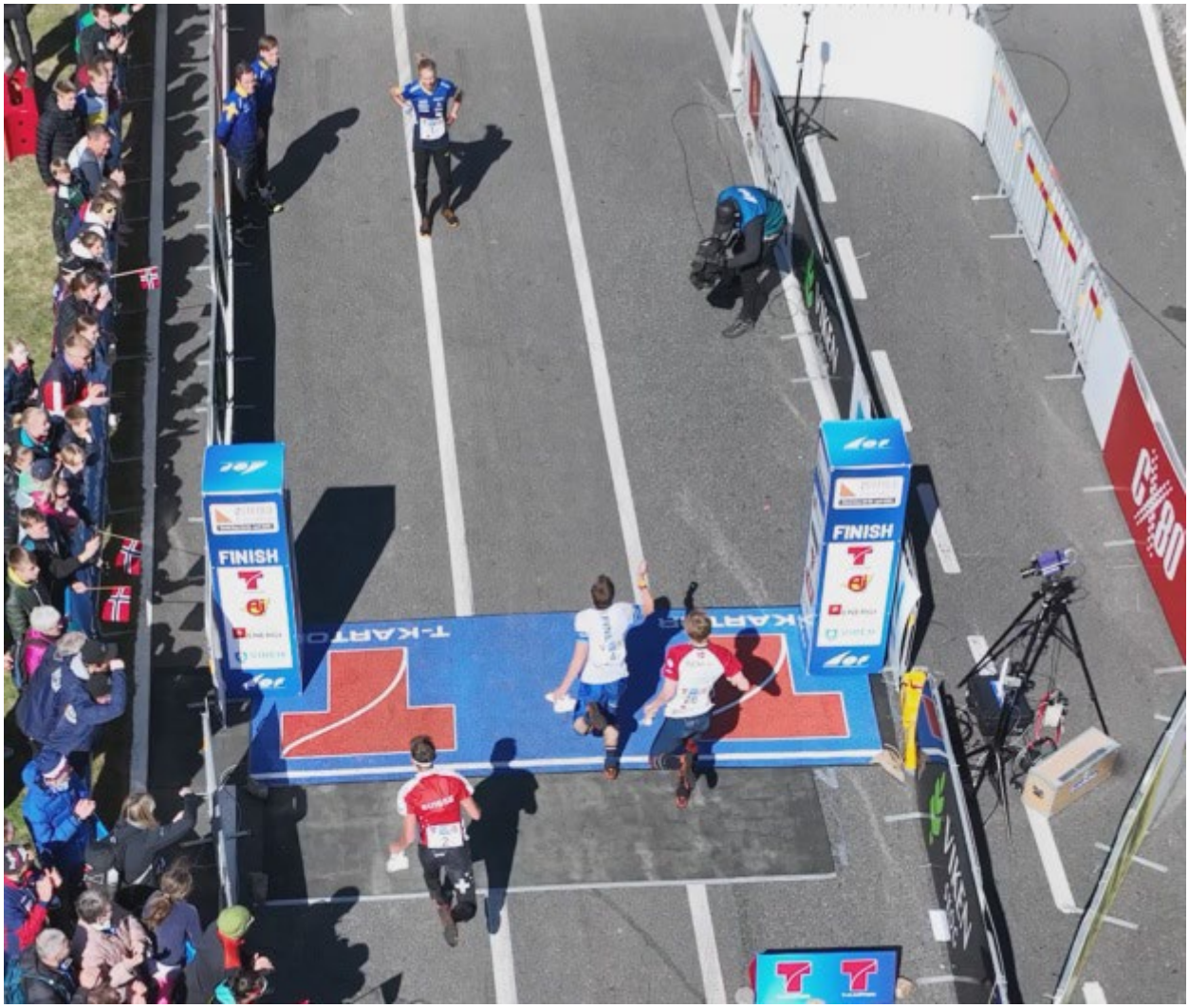
- The exact start time should be recorded by a *start gate.* The start clock may be separate from the finish clock, and the time of day for each athlete may be transmitted by radio. The clocks must be synced properly before the event.
- The finish time must be recorded by either a *photocell* or a *photo finish system*, and may additionally be recorded by an active transponder system with loop.
- The result is calculated by subtracting the recorded start time from the recorded finish time in full accuracy, and the result shall be truncated (rounded down) to the nearest whole second.
- The photocell may trigger two independent clocks for redundancy

4.0.1 First-to-finish races: Relays, chase starts and knock-out sprint

In these races, the rankings are determined by the order that the competitors cross the finish line. The times are of interest, but do not need to be completely accurate. The times may need to be adjusted to separate and correctly order competitors who finished close together. Using 1/10 precision in the results will minimize the number of manual adjustments that need to be made.

However, if two competitors record the same time (with 1/10 precision) but were separated by the finish judge and the photo finish, the time of one competitor shall be adjusted by 1/10 second. *Two competitors must only have the same placing and equal times if the finish judge and the photo finish are unable to separate the two competitors.*

- The start time may be recorded with an electronic start gun
- The finish time must be recorded by a *photocell*, a *photo finish system*, or an *active transponder triggered by a ground loop*
- A photo finish camera must be used to decide the rankings when the difference in time from photocell/transponder is less than 0.2s, and the time must be adjusted accordingly
- There must be a procedure for getting the photo finish picture to the TV production within 30 seconds
- The result list may use 1/10 precision. For Sprint Relay and Knock-out Sprint it is mandatory during World Orienteering Championships and Orienteering World Cup 2023.



Example of a tight sprint for the podium places at a World Cup Forest Sprint Relay.

4.1 Missed times

The times must be recorded by two systems with similar accuracy, system A and system B.

In cases when a single time is missing in system A, it should be taken from the system B.

If there is a systematic difference between the systems, the time taken from system B should be adjusted by a correction factor. The correction factor should be the difference between the time taken by the system B and the times of system A of the 3 competitors before the missed time.

The sum of the 3 times differences is divided by 3 which must be applied to the system B time.

5 Equipment

5.0 Timing equipment

Any equipment that is on the International Ski Federation (FIS) Homologated Timing Equipment[1] list fulfils the criteria for orienteering.

The follow timing equipment are within the specification for 0.1s precision, and have been used successfully in orienteering:

- Microgate REI2
- Microgate Racetime2
- Alge Timy-series (Timy, Timy 2 XE, PXE, Timy3)
- Emit ETS2
- Tag Heuer (ITS Chrono) CP5xx (Chronoprinter) – series
- SI Sprinter Station

Some examples of professional start gates:

- Alge STS
- Microgate Startgate
- TagHeuer (ITS Chrono) HL7

Photocells;

- Alge
- Microgate
- Tag Heuer (ITS Chrono) HL2

This list is not exhaustive, and other clocks and gates may be approved if they have sufficient documentation of precision and drift. The drift must be constant/stable during the race. If separate clocks are used on the start and at the finish the combined drift during the race must be less (more accurate) than the accuracy requirement.

Active transponder systems may be used to identify the athlete, and for timing as specified. However, the timing must be validated with the photocell or photo finish timing.

The following transponder systems are approved:

- Emit Emitag with the transponder on the arm/hand with a double antenna loop
- Sportident Air+ is allowed with a ground antenna loop

[1]As of January 2023 the latest version is v6.0 available from this page: https://assets.fis-ski.com/image/upload/v1636209117/fis-prod/assets/document-library/data-timing/timing-cross-country-nordic-combined.pdf

Transponders

Organisers shall provide the primary transponder in all televised World Orienteering Championship and Orienteering World Cup races. It is also recommended that the second transponder is provided by the Organisers, but it may also be provided by the athletes themselves.*

The vendors' instructions on setup, configuration and verification must be followed. As there are several ways to set up transponder timing wrongly, special attention must be paid to the equipment and materials involved in the arena construction around the antennas. The placement of metal (Iron) fences can seriously impact the characteristics of the antennas involved at the finish line. The installation must be verified before the race. Other vendors may be accepted, by the IT Commission.

*When Sportident Air+ is used, the production date of the built battery in the primary transponder should be newer than 2 years.

5.1 First-to-finish races – Photo finish

In races with live tv broadcast the photo finish system must use software and have routines that allows for a correct decision within 2 minutes after the athlete finishes.

The TV broadcaster wants the capability of sending a still image of the photo finish to the graphics operator within 30 seconds.

In practice a professional photo finish system is necessary, like FinishLynx, used at many WOCs (2003, 2010, 2014, 2019) or TagHeuer (ITS Chrono) ID-vision used at EOC 2018. These systems combine a high frame rate (50 fps – up to 2000 fps) with a software solution that enables fast and correct identification of the finish order. However, having the technical solution available is not enough – the operator must be familiar in how to operate the system.

The organiser should define who makes the final decision ("finish judge" §23.9) before the race starts.

At WOC 2019, in addition to FinishLynx, the regular video finish system (two security cameras), the apps Coach's Eye and SprintTimer were recording for backup at the relay events.

5.2 Software

It is common to use different software for timekeeping and integration with the tv graphics, and for punch control. This manual does currently not recommend any particular software, but the solution must of course have been thoroughly tested in other races before any high-level event.

6 Example setups

These are some examples of setups that has been proven and used in numerous international orienteering events. The vendors can provide further details on the hardware setup, training courses and instructions on how to use their equipment.

6.0 Individual finals

Role	Equipment	SportIdent	Emit	Purpose
Start	Start gate	Start gate + Sprinter Station SIAC control connected to start gate	Emit ETS2 + start gate	Used to determine exact start time
Finish, system A	Photocell	Sprinter Station + photocell	Emit ETS2 + photocell	System A
Finish, system B	Transponder finish	BS11 with ground loop antenna.	Emit ETS2 + dual loop	System B

An even better setup is to connect the photocell to two active result systems, so that the transponder is system C.

Regardless of which technology used for system A and B, it is good practice to have a 3rd and 4th level of backup as well, by recording every finished with a push-button on a timer with a printer, or alternatively using an iPad based program like Webscorer, and video preferably both from the side and the front of the finish line.

6.1 First-to-finish

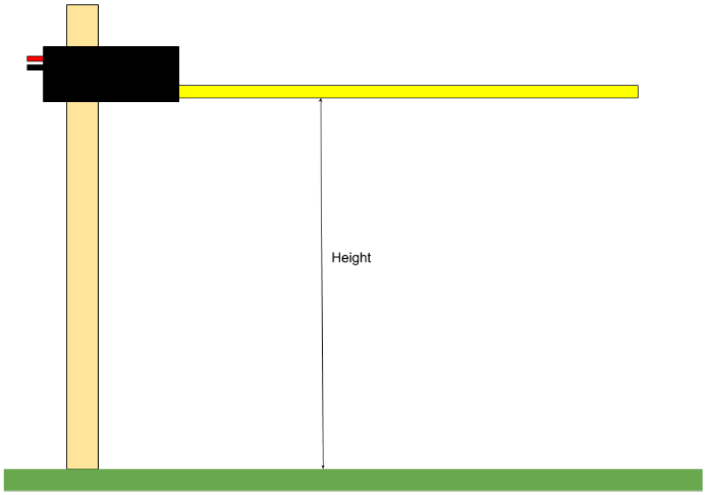
Role	Equipment	SportIdent	Emit	Purpose
Finish, system A	Transponder finish	BS11 with ground loop antenna.	Emit ETS2 + dual loop	System A
Finish, system B	Photo finish	FinishLynx / ITS Chrono	FinishLynx / ITS Chrono	System B. To help separate runners that are close

7 Hardware setup

7.0 Start gate setup

The start gate mounting post must be firmly affixed to the ground or connected to a fixed structure.

The height above the ground of the Start Gate must be so that athletes hit the bar and won't step over it. Recommended height is between 35 and 65 cm. Below is example of Start setup.



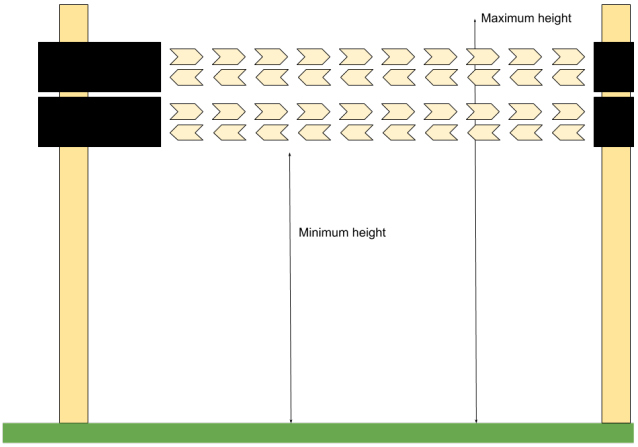
7.1 Photocell setup

The photocells must be installed in such way they are firmly affixed to the ground, and so that no athlete will strike the installation. The finish line must be as level as possible, and in such way that no athlete should be able to slide under or jump over the beam.

In foot-orienteering the photocell must be mounted so that the measuring point is at a minimum height of 50 cm and at a maximum 125 cm above ground. If photocells are used, both must be mounted within this limit.

We recommend two photo cells connected in serial, so that both beams must be triggered at the same time. They should be mounted at 80 cm and 120 cm above ground. This reduces the impact of runners “waving their hand”, and also reduces the number of false triggers.

Below is example of setups with photocells:



8 Terms and definitions

Start gate	A start gate has a bar connected to one or more switches that trigger the start clock(s).
Photocell	A photocell is used to create an invisible line with light that triggers the clock when an object crosses the line.
Photo finish system	A photo finish system consists of a line scan camera that records the finish line, together with the necessary software. This can be used to determine the finish time exactly, and separate athletes that crosses the finish line more or less at the same time (tie determination). This is important in first-to-finish races.
First-to-finish race	A race where the first athlete crossing the finish line is the winner, e.g. chase starts, knock-out sprints, relays and mass starts.
Video finish camera	A video recording device recording the athletes crossing the finish line, to determine the finish order, and finish time in cases where it is used as a backup system.
Manual hand timing (push-button)	The time is recorded by a timing device (clock) printing the time to paper or other medium and is triggered by a person who manually pushes a button when the athlete crosses the finish line. This system will work regardless of power supply.
Transponder	A transponder is a chip worn by the athlete, or fixed to the bike, which can be activated by an antenna, and be used for identification and timing.
Accuracy	The degree to which the result of a measurement, calculation, or specification conforms to the correct value or a standard. This is the “trueness” of the time published.

Precision	Refinement in a measurement, as represented by the number of digits given. This is the number of decimals used in the time published.
Truncation	The truncation to a full second must be done after the calculation of the race time. The decimals are thrown away. Example: - Start time: 10:01:00.10 - Finish time: 10:32:34.64 - Calculated race time: 31:34.54 - Official time: 31:34
System A System B	Two independent timekeeping systems, a primary (system A) - and a secondary (system B), shall be used continuously throughout the competition. They must have similar accuracy.